

Day 7 - IPv4 Addressing (pt 1)

Saturday, August 22, 2026 2:50 PM

7. IPv4 ADDRESSING : PART 1

OSI MODEL - NETWORK LAYER (Layer 3)

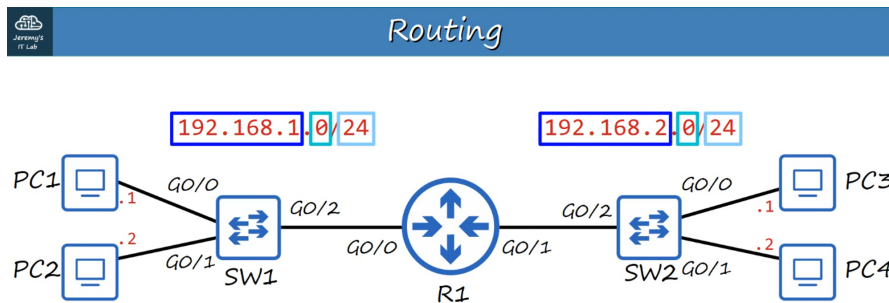
- Provides connectivity between end hosts on DIFFERENT networks (ie: outside of the LAN)
- Provides logical addressing (IP addresses)
- Provides path selection between SOURCE and DESTINATION
- ROUTERS operate at LAYER 3

ROUTING

SWITCHES (Layer 2 Devices) do not separate different networks. They connect and EXPAND networks within the same LAN.

By adding a ROUTER, however, between two SWITCHES, you create a SPLIT in the network; each with its own network IP address.

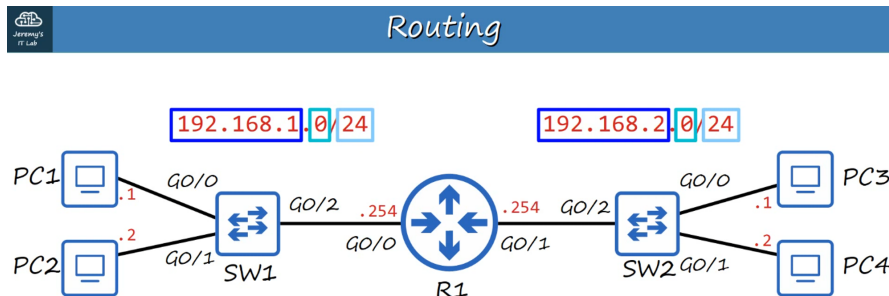
Example: 192.168.1.0/24 (255.255.255.0) 192.168.2.0/24 (255.255.255.0)



ROUTERS have unique IP Addresses for EACH of their interface connections, depending on their location.

The IP Address for the ROUTER's G0/0 Interface is: 192.168.1.254/24

The IP Address for the ROUTER's G0/1 Interface is: 192.168.2.254/24



The IP Address depends on network address of the LAN it is connected to.

The NETWORK portion of given IP Address will be the same for all HOSTS on a given LAN.

Example:

192.168.1.100 192.168.1.101 192.168.1.102 192.168.1.103 192.168.1.104 192.168.1.105 192.168.1.106 192.168.1.107 192.168.1.108 192.168.1.109 192.168.1.110 192.168.1.111 192.168.1.112 192.168.1.113 192.168.1.114 192.168.1.115 192.168.1.116 192.168.1.117 192.168.1.118 192.168.1.119 192.168.1.120 192.168.1.121 192.168.1.122 192.168.1.123 192.168.1.124 192.168.1.125 192.168.1.126 192.168.1.127 192.168.1.128 192.168.1.129 192.168.1.130 192.168.1.131 192.168.1.132 192.168.1.133 192.168.1.134 192.168.1.135 192.168.1.136 192.168.1.137 192.168.1.138 192.168.1.139 192.168.1.140 192.168.1.141 192.168.1.142 192.168.1.143 192.168.1.144 192.168.1.145 192.168.1.146 192.168.1.147 192.168.1.148 192.168.1.149 192.168.1.150 192.168.1.151 192.168.1.152 192.168.1.153 192.168.1.154 192.168.1.155 192.168.1.156 192.168.1.157 192.168.1.158 192.168.1.159 192.168.1.160 192.168.1.161 192.168.1.162 192.168.1.163 192.168.1.164 192.168.1.165 192.168.1.166 192.168.1.167 192.168.1.168 192.168.1.169 192.168.1.170 192.168.1.171 192.168.1.172 192.168.1.173 192.168.1.174 192.168.1.175 192.168.1.176 192.168.1.177 192.168.1.178 192.168.1.179 192.168.1.180 192.168.1.181 192.168.1.182 192.168.1.183 192.168.1.184 192.168.1.185 192.168.1.186 192.168.1.187 192.168.1.188 192.168.1.189 192.168.1.190 192.168.1.191 192.168.1.192 192.168.1.193 192.168.1.194 192.168.1.195 192.168.1.196 192.168.1.197 192.168.1.198 192.168.1.199 192.168.1.200 192.168.1.201 192.168.1.202 192.168.1.203 192.168.1.204 192.168.1.205 192.168.1.206 192.168.1.207 192.168.1.208 192.168.1.209 192.168.1.210 192.168.1.211 192.168.1.212 192.168.1.213 192.168.1.214 192.168.1.215 192.168.1.216 192.168.1.217 192.168.1.218 192.168.1.219 192.168.1.220 192.168.1.221 192.168.1.222 192.168.1.223 192.168.1.224 192.168.1.225 192.168.1.226 192.168.1.227 192.168.1.228 192.168.1.229 192.168.1.230 192.168.1.231 192.168.1.232 192.168.1.233 192.168.1.234 192.168.1.235 192.168.1.236 192.168.1.237 192.168.1.238 192.168.1.239 192.168.1.240 192.168.1.241 192.168.1.242 192.168.1.243 192.168.1.244 192.168.1.245 192.168.1.246 192.168.1.247 192.168.1.248 192.168.1.249 192.168.1.250 192.168.1.251 192.168.1.252 192.168.1.253 192.168.1.254 192.168.1.255

192.168.1.100 192.168.1.105 192.168.1.205

All of these addresses are on the SAME Network because the NETWORK PORTION of their IP Address is the same (192.168.1) while the HOST part (100,105,205) is UNIQUE!

When a BROADCAST message hits a ROUTER, it does NOT continue onward. It stays within the LOCAL LAN (Switch/Hosts).

IPv4 HEADER



IPv4 Header

IPv4 Header Format

Offsets	Octet	0								1								2								3							
Octet	Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	Version				IHL				DSCP				ECN				Total Length															
4	32	Identification								Flags				Fragment Offset																			
8	64	Time To Live												Protocol				Header Checksum															
12	96	Source IP Address																															
16	128	Destination IP Address																															
20	160	Options (if IHL > 5)																															
24	192																																
28	224																																
32	256																																

IP (or Internet Protocol) is the primary Layer 3 protocol in use today. Version 4 is the version in use in most networks.

IPv4 Headers contain MORE fields than the ETHERNET header.

IPv4 Headers contain a SOURCE IP Address and DESTINATION IP Address field.

This FIELD is 32-bits(4-bytes) in length (0-31)

192.168.1.254 (each decimal number represents 8 bits)

Translated to Binary:

11000000 . 10101000 . 00000001 . 11111110

EACH of these 8 bit groups are referred to as an OCTET

Since Binary is difficult to read for people, we use the Dotted Decimal format.

REVIEW of DECIMAL and HEXADECIMAL

Decimal & Hexadecimal				
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Decimal
(base 10)

3
3 * 1000

2
2 * 100

9
9 * 10

4
4 * 1

Hexadecimal (base 16)	<i>C</i>	<i>D</i>	<i>E</i>
	C * 256 (C = 12)	D * 16 (D = 13)	E * 1 (E = 14)
	3072	208	14

Decimal (base 10)

Ex: $3294 = (3 * 1000) + (2 * 100) + (9 * 10) + (4 * 1)$

Hexadecimal (base 16)

Ex: 3294, would be CDE

C (C * 256 / 12 * 256 = 3072) // 256ths position

D (D * 16 / D=13 so 16*13 = 208) // 16ths position

E (E * 1 / E = 14) // 1s position

Adding these up, we get 3294

So, how do we convert a BINARY NUMBER to a DECIMAL NUMBER? The same way we convert to Hexadecimal.

10001111

So:

1 * 128 = 128

1 * 8 = 8

1 * 4 = 4

1 * 2 = 2

1 * 1 = 1

Add them all up : $128 + 8 + 4 + 2 + 1 = 143$

The answer is 143.

Another example:

01110110

1 * 64 = 64

1 * 32 = 32

1 * 16 = 16

1 * 4 = 4

1 * 2 = 2

Add them all up: $64 + 32 + 16 + 4 + 2 = 118$

The answer is 118.

Another example:

11101100

1 * 128 = 128

1 * 64 = 64

1 * 32 = 32

1 * 8 = 8

$$1 * 4 = 4$$

Add them all up: $128 + 64 + 32 + 8 + 4 = 236$

The answer is 236.

So, how do we convert a DECIMAL NUMBER to a BINARY NUMBER?

Take the number 221.

We can take that number and start subtracting it from LEFT to RIGHT of our Binary slots.

221

$221 - 128 = 93$ so we place a 1 in the "128" slot

10000000

$93 - 64 = 29$ so we place another 1 in the "64" slot

$29 - 32$ isn't possible so we place a 0 in the "32" slot

$29 - 16 = 13$ so we place a 1 in the "16" slot

$13 - 8 = 5$ so we place a 1 in the "8" slot

$5 - 4 = 1$ so we place a 1 in the "4" slot

$1 - 2$ isn't possible so we put a 0 in the "2" slot

$1 - 1$ is possible so we put a 1 in the "1" slot

This, then, allows us to write out the BINARY number for 221.

It is : 11011101

Another example: 127

$127 - 128$ is not possible so 0 in "128"

$127 - 64$ is possible so 1 in "64"

$63 - 32$ is possible so 1 in "32"

$31 - 16$ is possible so 1 in "16"

$15 - 8$ is possible so 1 in "8"

$7 - 4$ is possible so 1 in "4"

$3 - 2$ is possible so 1 in "2"

1 is possible so 1 in "1"

So 127, in BINARY, is 0111 1111

Another example: 207

Alternatively, you can subtract the number from '255' (which is 1111111). The remainder, then, can be used to "find" where the 0's are in the binary number.

$255 - 207 = 48$ so ...

1 1 0 0 1 1 1 1 ($32 + 16 = 48$)

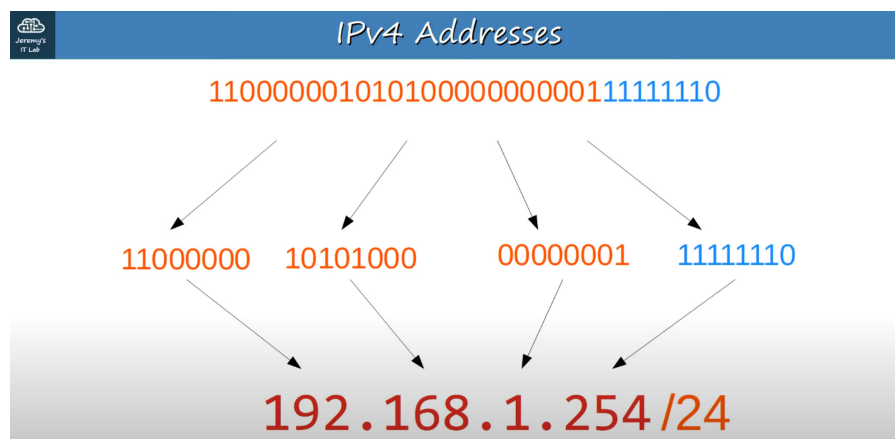
11001111 is the correct answer.

IPv4 ADDRESSES

So we now know that IP Addresses are the Dotted Decimal conversion of a series of BINARY NUMBERS (broken up into 4 OCTETS) like so:

192.168.1.254/24

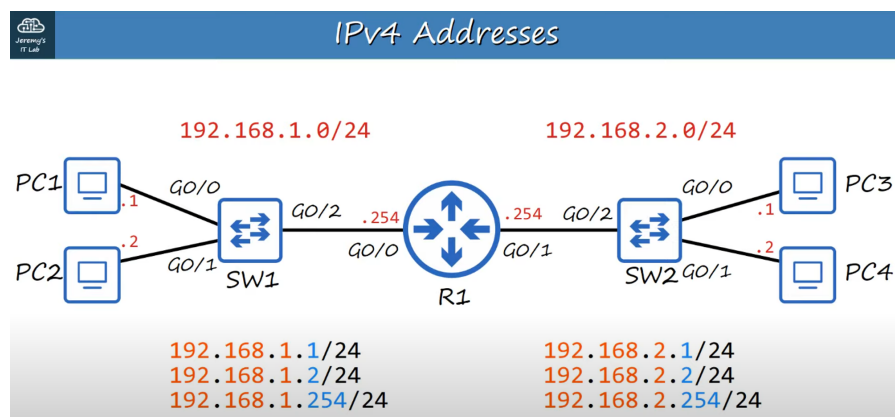
But what does the /24 stand for?



It means the FIRST 24 BITS of this address represent the NETWORK portion of the address.

192.168.1 is the NETWORK PORTION (the first 3 OCTETS)

.254 is the HOST PORTION (the last OCTET)



CONVERT this BINARY number into an IPv4 Address:

10011010010011100110111100100000

10011010 . 01001110 . 01101111 . 00100000

Octets:

1. $128 + 16 + 8 + 2 = 154$
2. $64 + 8 + 4 + 2 = 78$
3. $64 + 32 + 8 + 4 + 2 + 1 = 111$
4. 32

The IPv4 address is: 154.78.111.32/16

154.78 is the NETWORK PORTION 111.32 is the HOST PORTION

Another Example:

00001100100000001111101100010111

00001100 . 10000000 . 11111011 . 00010111

Octets:

1. $8 + 4 = 12$
2. 128
3. $255 - 4 = 251$
4. $16 + 4 + 2 + 1 = 23$

The IPv4 address is: 12.128.251.23/8

12 is the NETWORK PORTION 128.251.23 is the HOST PORTION

IPv4 ADDRESS CLASSES

IPv4 ADDRESSES are split up into 5 different 'classes'. The class of an IPv4 is determined by the FIRST OCTET of the address.

CLASS FIRST OCTET FIRST OCTET NUMERIC RANGE

A 0xxxxxxx 0-126 + 127 'loopback' B 10xxxxxx 128-191 C 110xxxxx 192-223 D 1110xxxx 224-239 E 1111xxxx 240-255

From the above chart, if the FIRST OCTET STARTS with 0, the numeric RANGE of possible first DOTTED DECIMAL is between 0-127.

The CLASSES we will be focusing on are CLASS A to CLASS C.



IPv4 Address Classes

Class	Leading bits	Size of network number bit field	Size of rest bit field	Number of networks	Addresses per network
Class A	0	8	24	128 (2^7)	16,777,216 (2^{24})
Class B	10	16	16	16,384 (2^{14})	65,536 (2^{16})
Class C	110	24	8	2,097,152 (2^{21})	256 (2^8)

D CLASS are reserved for 'MULTICAST' ADDRESSES

E CLASS are reserved for 'EXPERIMENTAL' ADDRESSES

A CLASS USUALLY have a range of 1-126? WHY?

Because 127 is usually reserved for 'loopback addresses'

127.0.0.0 to 127.255.255.255 are used to test the network.

- Used to test the 'Network stack' (OSI & TCP/IP model) on the local device.

IPv4 Address Classes

Class	First octet	First octet numeric range	Prefix Length
A	0xxxxxxx	0-127	/8
B	10xxxxxx	128-191	/16
C	110xxxxx	192-223	/24

The PREFIX LENGTH is the LENGTH of the NETWORK PORTION of the Address.

From the examples above:

12.128.251.23/8 is a CLASS A Address 154.78.111.32/16 is a CLASS B Address 192.168.1.254/24 is a CLASS C Address

Because the NETWORK portion of CLASS A is so short, it means there are a LOT more potential Hosts.

Because the NETWORK portion of CLASS C is so long, it means fewer potential Hosts.

NETMASK

Netmask

Class A: /8 255.0.0.0
(11111111 00000000 00000000 00000000)

Class B: /16 255.255.0.0
(11111111 11111111 00000000 00000000)

Class C: /24 255.255.255.0
(11111111 11111111 11111111 00000000)

A NETMASK is written like a Dotted Decimal IP Address

CLASS A: /8 = 255.0.0.0

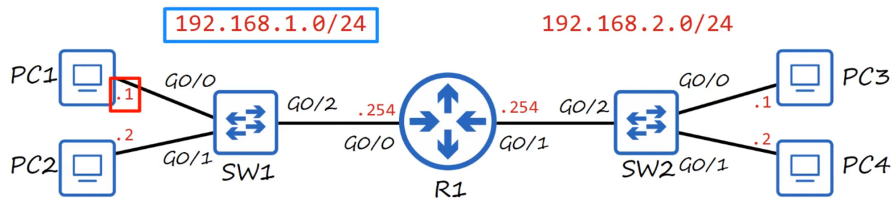
CLASS B: / 16 = 255.255.0.0

CLASS C: /24 = 255.255.255.0

NETWORK ADDRESSES



Network Address



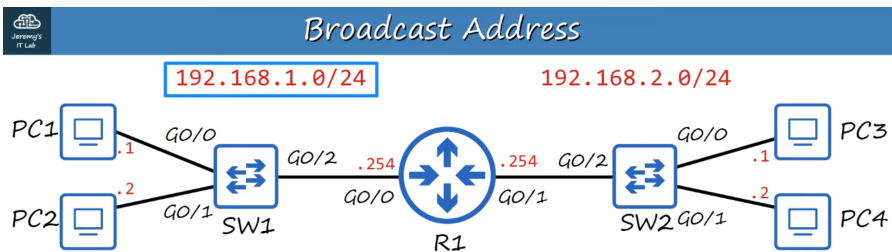
Host portion of the address is all 0's = Network Address

The network address CANNOT be assigned to a host.

If the HOST PORTION of an IP ADDRESS is ALL 0's, it means it is the NETWORK ADDRESS = the identifier of the network itself.

Example: 192.168.1.0/24 = THIS is a NETWORK ADDRESS.

A NETWORK ADDRESS cannot be assigned to a HOST. A NETWORK ADDRESS is the FIRST ADDRESS.



Host portion of the address is all 1's = Broadcast Address

The broadcast address CANNOT be assigned to a host.

If the HOST PORTION of an IP ADDRESS is ALL 1's, it means it is the BROADCAST ADDRESS for the network.

A BROADCAST ADDRESS cannot be assigned to a HOST.

DESTINATION IP : 192.168.1.255 (Broadcast IP address) DESTINATION MAC : FFFF.FFFF.FFFF (Broadcast MAC address)

Because of the two 'reserved' addresses, the range of USABLE HOST ADDRESSES is 1 to 254.